



The Butterfly in the Classroom

AI Tools for Education | *අධ්‍යාපන අංශයේ පාඨමාලා* *Academic Context*

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60-minute session | NIBM School of Computing and Engineering

The Same Student. Three Different Worlds.

Version A

Handwritten draft

Photosynthesis is a proseses plants do with sunlight. They take in CO2 and water and make energy. Its important for all living things because of the food chain and stuff.

Version B

AI-polished, undisclosed

Photosynthesis is the biological process by which plants convert light energy into chemical energy. Utilising chlorophyll, they absorb carbon dioxide and water to produce glucose, releasing oxygen as a byproduct.

Version C

AI guided by lecturer framing

Photosynthesis is the plant's own food factory. Sunlight powers the line. CO2 and water are the inputs. Glucose is the product. Oxygen is the waste that keeps us alive.

Which one is the integrity problem?

“The problem is not the tool.

It is the absence of intentional pedagogy around it.

We have been asking the wrong question.

Today we ask the right one.

One Frame for the Next 60 Minutes



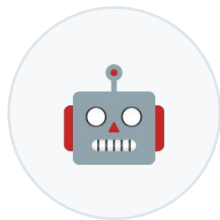
AI = Thinking Partner

Not an answer machine.

A tool that amplifies the quality of your questions.
The output is a direct function of the care you bring to the input.

Today's demonstration will prove this. Literally.

Three Families, One Decision



Generative AI

Claude, ChatGPT, Gemini

Creates new text from your instructions.

Every output is unique.

Exquisitely sensitive to how you phrase the prompt.



Productivity Wrappers

Grammarly, Canva AI, Turnitin

Enhances work that already exists.

Lower creative variance.

Narrower scope, lower academic risk.



Research Assistants

Elicit, Consensus, Perplexity

Finds and synthesises sources.

Useful for literature scoping.

Not reliable for generating citations.

What You Can Use Tomorrow

✓ Free, No VPN Required

ChatGPT Claude.ai Gemini Perplexity Canva AI

Works on any Sri Lankan broadband or Wi-Fi connection

🔑 Institutional Licence Needed

Turnitin AI features ChatGPT Team Microsoft Copilot

Ask your library whether NIBM holds an active seat

⚠ Mobile Data Caution

Any LLM on 4G or prepaid data

Long responses stream heavily -- use Wi-Fi for live demos

Not a Better Search Engine



SEARCH

Google finds a document that already exists.

It retrieves.

Phrasing changes the ranking.
Not the answer itself.

VS



GENERATE

Claude writes a new document from scratch every time.

It generates.

Phrasing changes the output fundamentally.

Retrieval is robust to phrasing. Generation is exquisitely sensitive to it.



4 MINUTES

Ask the AI to explain a topic from your own subject to a bright 18-year-old who has never studied it.

1

Open Claude.ai
or Gemini

2

Type your prompt
exactly as it comes

3

Read every word
of the output

One Clause Changed Everything

PROMPT A

Explain photosynthesis to a first year university student.

OUTPUT

Photosynthesis is the biological process by which plants convert light energy into chemical energy. Using chlorophyll, they absorb carbon dioxide and water to produce glucose, releasing oxygen as a byproduct. The process has two stages: light-dependent reactions in the thylakoid, and the Calvin cycle in the stroma.



PROMPT B

Explain photosynthesis to a first year university student who finds science abstract and dull.

OUTPUT

Think of a plant as a solar-powered kitchen. Sunlight is the electricity. Carbon dioxide and water are the raw ingredients it orders in. Glucose -- the meal it cooks -- is what keeps it alive. The byproduct it vents out the window? That is the oxygen you are breathing right now.

This is sensitive dependence on initial conditions.

Three Questions to Hold

1

What did the AI get right?

Accuracy check -- your discipline knowledge is the filter

2

What was wrong or too shallow?

Gap check -- where did phrasing or scope let it down?

3

What would have taken you longer to write yourself?

Value check -- where is the legitimate time saving?

You as Designer



Generate a marking rubric at a specified Bloom's level



Draft a case study for your own subject area



Simplify a reading passage for ESL (English as a Second Language) students



Produce 20 exam questions mapped to your learning outcomes

Prompt: "Write a Bloom's Level 4 rubric for a 2000-word essay on data breach response. Five criteria. Table format." -- Done in 8 seconds.

You as Assessor



Generate first-draft marking criteria for a new assignment



Draft personalised feedback from your own brief notes



Check marking consistency across a script sample



Translate verbal comments into written feedback language

You remain professionally accountable for every word that leaves under your name.

The Alarm That Cries Wolf



The Turnitin AI Detection Problem

- AI detection tools produce significant false positives on non-native English writing
- Sri Lankan students writing formal academic English are disproportionately flagged
- A high AI score is not evidence of AI use -- it is a signal to open a conversation
- No detection tool should trigger a sanction without human review and dialogue

An alarm that fires 40% of the time on non-fires is not a reliable alarm.

You as Researcher



Scope a literature field before you search databases



Stress-test your argument structure before you write



Draft grant application language from bullet points



Synthesise 10 abstracts into a thematic overview draft

The next slide explains why every source AI names must be verified independently.

This Reference Does Not Exist.

Wijesinghe, R. K., & Perera, L. (2023). Integrating digital tools in Sri Lankan tertiary classrooms: A systematic review. Journal of South Asian Educational Research, 11(3), 14-22. DOI: 10.1177/jsaer.2023.0311

FABRICATED

AI language models do not retrieve. They predict. Predictions look exactly like facts.

The journal name, volume, page numbers, authors, and DOI are all plausible. None of them exist.

Would you have caught it?

The Questions Your Faculty Needs to Own

1

Does your faculty have an AI disclosure policy?

2

If a student substantially directs AI to produce work, is that plagiarism?

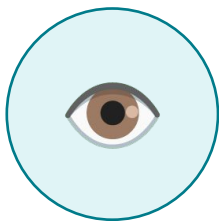
3

If you use AI to draft feedback, are you still professionally accountable for it?

These are not rhetorical.

A Policy Seed

What responsible use looks like before a formal policy exists:



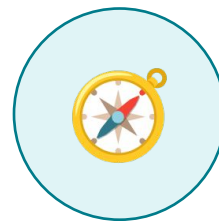
Transparency over Secrecy

Disclose when AI assisted your work. Students model what lecturers normalise. If you use it and hide it, so will they.



Accountability with Acknowledgement

AI-assisted work is still your work. Errors and tone are your responsibility. Acknowledge the tool; own the output.



Guidance over Prohibition

Blanket bans create adversarial relationships. Clear guidance on what is permitted and why creates professional culture.



The Wing Has Not Flapped Yet

Small decisions made in faculty meetings this year will cascade into institutional culture in three years.

Most Sri Lankan HEIs have not yet drawn this line.

This room can be the butterfly.

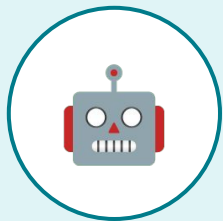
Your Initial Condition

Write it. Say it out loud. It matters more than any summary.

**One thing I will try
before next month...**

**One concern I
still carry...**

Three Starting Points



Claude.ai

claude.ai

Free tier. Strong for rubric drafting, feedback generation, and lesson planning. Start here.



Elicit.org

elicit.org

Free for researchers. Searches academic literature and extracts key findings. Reliable for systematic scoping -- do not use for citation lists.



EDUCAUSE AI Resources

edUCAUSE.edu / AI focus

Policy frameworks, case studies, and guidance for HEI leaders and teaching faculty. Well-grounded in institutional realities.



The initial condition has already changed.



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